

Presence on the face

This is done and automated. It was going to be a hand-wiring job in the animation graph; it is not, and this document is now a description rather than a set of instructions. Kept because the *why* is worth having when it next needs changing, and because the failure modes are still real.

```
Scripts/wire_presence_to_face.py      # reparent the face blueprint, set curve names
Scripts/add_presence_component.py     # attach presence to the character
Scripts/bind_face_blueprint.py        # point the face at ABP_AmandaFace
```

All three must run after any character rebuild: BP_MHC_Seo and its face live under /Game/MetaHumans, which assemble_metahuman.py regenerates and git ignores.

How it works

The presence layer - blink, gaze, breath, drift - runs as UAmandaPresenceComponent on the character, ticking and subscribed to the orchestrator's bridge. It is a port of avatar-orchestrator/src/amanda/presence/schedulers.py, which is the reference implementation and holds the reasoning behind every constant.

Getting those numbers onto the rig looked like animation-graph work, which cannot be scripted: graphs are authored by hand and Python cannot reach them. It isn't. UAnimInstance::AddCurveValue sets a named curve on the evaluated pose directly, so UAmandaFaceAnimInstance - a native parent class for ABP_AmandaFace - writes presence into the rig every frame with no nodes wired at all. The blueprint keeps every graph Epic shipped in it; only its parent changed.

NativeUpdateAnimation runs before the graph evaluates and the curves it sets apply to the result, so presence layers *after* the Live Link pose. The mouth is untouched: nothing here writes a jaw or lip curve, and nothing should. The mouth is solved from audio by MetaHuman and presence has no business overriding it.

The curves it drives

Not guesses. UAmandaFaceCurveLibrary::ListCurveNames reports 2038 curves on this rig; these are the eight that matter. The rig also carries ARKit aliases (EyeBlinkLeft), but the CTRL_expressions_* controls are what the face board actually drives.

Purpose	Curves
Blink	CTRL_expressions_eyeBlinkL, CTRL_expressions_eyeBlinkR
Look left	CTRL_expressions_eyeLookLeftL, CTRL_expressions_eyeLookLeftR
Look right	CTRL_expressions_eyeLookRightL, CTRL_expressions_eyeLookRightR
Look up	CTRL_expressions_eyeLookUpL, CTRL_expressions_eyeLookUpR
Look down	CTRL_expressions_eyeLookDownL, CTRL_expressions_eyeLookDownR

They are configuration, not constants, on the blueprint's class defaults - a different MetaHuman may name them differently. To find out what a rig offers:

```
unreal.AmandaFaceCurveLibrary.list_curve_names(mesh, "eyeBlink")
```

LidOpen is 1.0 for **open**; blink curves want the inverse, and the anim instance does that conversion. If she ever sits with her eyes serenely shut, that inversion is the first place to look.

Verifying

Animation blueprints do not evaluate in the editor viewport, so **Simulate or Play**, never the viewport. Verifying by eye in the viewport cost an afternoon here.

```
Scripts/simulate_and_shoot.py    # burst of frames while the world ticks
```

Then measure rather than squint - the eye region against the median frame. Last run, 24 frames: baseline 0.7-1.0, one frame at **6.53** (a blink caught mid-closure) and a seven-frame plateau at ~3.9 (a held gaze shift). Presence sampled directly over 40 seconds: 14 blinks (21/min, humans do 10-20), lid range 0.000-1.000, eye yaw -12.3 deg to +11.9 deg, all five gaze targets visited.

What is not done

Brows and the rest of the performance. The director's preset and intensity reach the component and stop there. FaceState.Preset and .Intensity are published and nothing consumes them, so the face has presence but no *expression*.

Breath, as far as the head shows it. The scheduler's own description is "chest and shoulders, plus a trace of it in the head". The chest and shoulders are out of scope - this is head and face cinematography - so what is wanted here is only that trace: the small vertical settle and release in the neck that says someone is breathing. FaceState.Breath is published and nothing reads it yet.

Head rotation. PresenceHeadRotation is computed and exposed but not applied; the head needs either the anim graph's ARKit_HeadRotation or a bone modifier, and neither is wired.

If it stops working

Nothing moves. Check you are in Simulate or Play. Then check bHasPresence on the anim instance - false means the component was not found on the actor.

It worked, then stopped after a rebuild. The three scripts at the top. /Game/MetaHumans is regenerated wholesale.

Eyes shut permanently. The LidOpen inversion.

Head twitches. Presence and Live Link both claiming it. LLink_Face_Head should be false if presence ever drives the head.

What the values mean

Field	Range	Notes
LidOpen	1.0 open to 0.0 closed	Blink curves take the inverse
EyeYaw / EyePitch	+/-16 deg	Includes micro-saccades; never perfectly still
HeadYaw / HeadPitch / HeadRoll	+/-3 deg	Gaze lag plus drift. Deliberately tiny
Breath	0.0 to 1.0	Chest, not face
Preset / Intensity	-	From the performance director. Unconsumed
GazeTarget	enum	Diagnostics

Change behaviour in `schedulers.py` first and port it, so the reference implementation and the renderer do not drift apart.